

GATE SMASHERS PLAYLIST EXAM MAPPING

Course Pattern: TE IT - 2024 Pattern (SPPU)
Playlist Channel: Gate Smashers (YouTube)

Academic Term: Semester V (2026)
Course Code: PCC-303-ITT

NOTE:

How to Use This Mapping: This document aligns the complete Gate Smashers Theory of Computation (TOC) YouTube playlist into the five core syllabus units of Savitribai Phule Pune University (SPPU).

Exam Preparation Note: Lectures highlighted in **bold** represent high-priority exam topics explicitly tested in SPPU end-semester examinations and fully detailed in the unit-wise theory notes (e.g., DFA Construction, Minimization, NFA to DFA, Epsilon NFA, Moore & Mealy Machines, Arden's Theorem, Closure Properties, CFG Derivation Trees, Pushdown Automata, Turing Machines, and Decidability).

Quick Statistics:

- **Total Playlist Videos Extracted:** 68 (Total Duration: 10h 35m)
- **Unit 1 (Finite Automata):** 27 Videos (4h 06m)
- **Unit 2 (Regular Expressions & Languages):** 14 Videos (2h 02m)
- **Unit 3 (Context-Free Grammars & Languages):** 13 Videos (1h 53m)
- **Unit 4 (Pushdown Automata & Turing Machines):** 9 Videos (1h 29m)
- **Unit 5 (Decidability & Computational Complexity):** 5 Videos (1h 03m)

Unit 1: Finite Automata (4h 06m)

Syllabus Topics Covered: Symbols, Alphabet, Strings, Formal Languages, DFA Construction, Transition Tables/Diagrams, Extended Transition Function, DFA Minimization, NFA, ϵ -NFA, Conversions (ϵ -NFA $\rightarrow$ NFA $\rightarrow$ DFA), Moore & Mealy Machines, Inter-conversions.

Lecture Topic / Title	Duration	Watch Link
Lec-1: Syllabus of TOC for GATE / NET / Imp Points	6:03	Play ↗
Lec-2: Introduction to TOC What is Language in TOC	12:21	Play ↗
Lec-3: What is Automata in TOC Theory of Computation	5:18	Play ↗
Lec-4: Power of Sigma Σ in TOC Kleene closure	8:31	Play ↗
Lec-5: What is Grammar in TOC Must Watch	11:08	Play ↗
Lec-6: What is DFA in TOC with examples	13:14	Play ↗
Lec-7: DFA Example 1 How to Construct DFA	8:12	Play ↗
Lec-8: DFA Example 2 Strings ending with 'a'	5:51	Play ↗
Lec-9: DFA starting with 'a' and ending with 'b'	8:34	Play ↗
Lec-10: DFA Not starting with 'a' OR Not ending with 'b'	7:24	Play ↗
Lec-11: DFA of binary strings divisible by 3	7:26	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-12: DFA 2nd symbol '0' and 4th symbol '1'	7:19	Play ↗
Lec-13: DFA for Even a and Even b Combination cases	9:35	Play ↗
Lec-14: Equivalence of DFA with examples	8:51	Play ↗
Lec-15: Minimization of DFA with example	17:00	Play ↗
Lec-16: Limitations and Applications of DFA	12:40	Play ↗
Lec-17: What is NFA in TOC Nondeterministic Automata	9:01	Play ↗
Lec-18: Design NFA where 2nd last bit is 1	6:09	Play ↗
Lec-19: DFA vs NFA with examples	7:57	Play ↗
Lec-20: Convert NFA to DFA with example	9:37	Play ↗
Lec-21: Epsilon NFA Formal Definition (ϵ-NFA)	7:55	Play ↗
Lec-22: Eliminate Epsilon moves (ϵ-NFA to NFA)	7:40	Play ↗
Lec-23: Conversion from Epsilon ϵ-NFA to DFA	10:46	Play ↗
Lec-24: Moore Machine in TOC with example	9:51	Play ↗
Lec-25: Mealy Machine in TOC Formal Definition	7:45	Play ↗
Lec-26: Difference between Mealy and Moore Machine	7:55	Play ↗
Lec-27: Moore to Mealy Conversion with example	8:46	Play ↗
Lec-28: Mealy to Moore Conversion with Example	12:11	Play ↗

Unit 2: Regular Expressions and Languages (2h 02m)

Syllabus Topics Covered: Definition and Identities of RE, Operators of RE, RE to FA conversion (Thompson's construction), Arden's Theorem & Proof, FA to RE conversion, Closure properties of RLs, Applications of RE.

Lecture Topic / Title	Duration	Watch Link
Lec-29: Regular Expressions in TOC with examples	9:59	Play ↗
Lec-30: Regular Expressions for Finite Languages	8:29	Play ↗
Lec-31: Regular Expressions for Infinite Languages	13:46	Play ↗
Lec-32: Important Questions on Regular Expressions	5:51	Play ↗
Lec-33: Pumping Lemma for regular languages	12:00	Play ↗
Lec-34: Closure properties of regular languages	9:33	Play ↗
Lec-35: Quotient operation in TOC with example	9:52	Play ↗
Lec-36: Reversal Operation in TOC	6:28	Play ↗
Lec-37: INIT Operation in TOC	5:50	Play ↗
Lec-38: Regular languages Not Closed under Infinite Union	3:06	Play ↗
Lec-39: Closure Properties Of Various Languages	12:33	Play ↗
Lec-40: Languages, Automata, Grammars Comparison	12:21	Play ↗
Lec-41: Homomorphism in Regular Languages	5:32	Play ↗

Lecture Topic / Title	Duration	Watch Link
Lec-42: Inverse Homomorphism in Regular Languages	7:13	Play ↗

Unit 3: Context-Free Grammars and Languages (1h 53m)

Syllabus Topics Covered: Chomsky Hierarchy, Regular Grammars (RG, LLG, RLG), Conversions ($LLG \leftrightarrow RLG$, $RG \leftrightarrow FA$), Context-Free Grammar (CFG) Definition, Derivation/Parse Trees, Leftmost/Rightmost Derivations, Ambiguity, CFG Simplification, CYK Algorithm.

Lecture Topic / Title	Duration	Watch Link
Lec-43: CFL and CFG Introduction & Syllabus	4:03	Play ↗
Lec-44: Closure Properties of CFL	8:57	Play ↗
Lec-45: Questions on DCFL and CFL	11:05	Play ↗
Lec-46: What is Context Free Grammar (CFG)	7:57	Play ↗
Lec-47: Convert CFL to CFG with examples	15:00	Play ↗
Lec-48: Derivation Tree / Parse Tree with example	6:33	Play ↗
Lec-49: Left Most & Right Most Derivation in CFG	6:22	Play ↗
Lec-50: Ambiguous vs Unambiguous Grammar	9:30	Play ↗
Lec-51: Recursive vs Non-Recursive CFG	5:44	Play ↗
Lec-52: Remove Unit Production from CFG	7:10	Play ↗
Lec-53: Remove Null Production from CFG	7:48	Play ↗
Lec-54: CYK Algorithm Membership Algorithm in CFG	17:00	Play ↗
Lec-55: CNF vs GNF Normal Forms for CFG	6:33	Play ↗

Unit 4: Pushdown Automata and Turing Machines (1h 29m)

Syllabus Topics Covered: Formal Definition of PDA, Instantaneous Description (ID), Acceptance by Final State vs Empty Stack, Design of PDA, Equivalence of CFG and PDA, Turing Machine (TM) Formal Definition, IDs, Language Acceptance, Design of TM, TM Variants, Multi-Tape TM, Universal Turing Machine.

Lecture Topic / Title	Duration	Watch Link
Lec-56: What is Pushdown Automata in TOC	10:58	Play ↗
Lec-57: Design PDA for $0^n 1^{2n}$ CFL Language	13:13	Play ↗
Lec-58: Design PDA for $\{wm \mid n_{a(w)} = n_{b(w)}\}$	11:25	Play ↗
Lec-59: What is Linear Bounded Automata (LBA)	4:37	Play ↗
Lec-60: Introduction to Turing Machine Definition	9:03	Play ↗
Lec-61: Design Turing Machine for $a^n b^n$	11:56	Play ↗
Lec-62: Design Turing Machine for $a^n b^n c^n$	11:18	Play ↗
Lec-63: Turing Machine for 1's Complement	8:06	Play ↗
Lec-64: Modifications and Variants in Turing Machine	9:20	Play ↗

Unit 5: Decidability and Computational Complexity (1h 03m)

Syllabus Topics Covered: Church-Turing Thesis, Recursive vs Recursively Enumerable Languages, Decidable Problems for Regular & CFL, Undecidability, Halting Problem, Pumping Lemma applications, Computational Complexity (P, NP, NP-Complete, NP-Hard, SAT, Cook's Theorem, Node Cover).

Lecture Topic / Title	Duration	Watch Link
Lec-65: Recursive vs Recursive Enumerable Languages	6:56	Play ↗
Lec-66: Decidability & Undecidability Table in TOC	7:57	Play ↗
Lec-67: Important Questions on Decidability & Closure	8:38	Play ↗
Lec-68: TOC Most Imp 10 Questions Part 1	31:00	Play ↗
Lec-69: TOC Most Imp 10 Questions Part 2	8:55	Play ↗

End of Theory of Computation Gate Smashers Playlist Mapping — SPPU TE IT 2024 Pattern